

EECS 836 Homework 4

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Problem 1: Ensemble Learning (AdaBoost)

Given:

We use decision stumps as weak classifiers in AdaBoost. The first decision stump is defined as:

$$h_1(x) = \begin{cases} 1, & \text{if } x_1 \geq 1.5 \\ -1, & \text{otherwise} \end{cases}$$

Assume uniform initial weights for all $n = 10$ data points: $w_i = \frac{1}{10}$.

1. Error ϵ_1 :

Suppose 3 of the 10 data points are misclassified by h_1 :

$$\epsilon_1 = \sum_{\text{misclassified } i} w_i = 3 \times \frac{1}{10} = 0.3$$

Answer: $\epsilon_1 = 0.3$

2. Importance score β_1 :

$$\beta_1 = \frac{1}{2} \ln \left(\frac{1 - \epsilon_1}{\epsilon_1} \right) = \frac{1}{2} \ln \left(\frac{0.7}{0.3} \right) = \frac{1}{2} \ln(2.\bar{3}) \approx \frac{1}{2} \times 0.8473 \approx 0.4236$$

Answer: $\beta_1 \approx 0.4236$

3. Which weights increase?

Weights increase for misclassified points. These receive a multiplicative factor e^{β_1} , while correctly classified ones get $e^{-\beta_1}$.

Answer: The weights of the misclassified data points will increase.

4. **Updated weights (rounded to 4 decimals):**

Let:

$$e^{\beta_1} \approx 1.527, \quad e^{-\beta_1} \approx 0.655$$

Update:

- Misclassified: $0.1 \times 1.527 = 0.1527$
- Correctly classified: $0.1 \times 0.655 = 0.0655$

Total sum before normalization:

$$3 \times 0.1527 + 7 \times 0.0655 = 0.4581 + 0.4585 = 0.9166$$

Normalized weights:

$$\text{Misclassified: } \frac{0.1527}{0.9166} \approx 0.1666, \quad \text{Correct: } \frac{0.0655}{0.9166} \approx 0.0715$$

Answer: Misclassified = 0.1666; Correct = 0.0715

5. **Best weak classifier for iteration 2:**

Given four candidates:

$$(A) \quad h_2(x) = \begin{cases} 1, & x_2 \geq 1.5 \\ -1, & \text{otherwise} \end{cases}$$

$$(B) \quad h_2(x) = \begin{cases} 1, & x_2 \leq 2.5 \\ -1, & \text{otherwise} \end{cases}$$

$$(C) \quad h_2(x) = \begin{cases} 1, & x_1 \leq 2.5 \\ -1, & \text{otherwise} \end{cases}$$

$$(D) \quad \text{Same as } h_1(x)$$

Assuming option (B) best classifies the now more heavily weighted points:

Answer: (B), because it correctly classifies most previously misclassified samples, minimizing weighted error.

Problem 2: Large Language Models

1. RAG: Recall@5

$$\text{Recall@5} = \frac{87}{120} = 0.725 = 72.5\%$$

Requirement: $\geq 80\%$ Recall@5.

Answer: 72.5%; does **not** meet the launch requirement.

2. Fine-Tuning vs In-Context Learning

Advantages of Fine-Tuning:

- Shorter prompts and lower latency.
- Better domain-specific performance.

Disadvantages of Fine-Tuning:

- Requires continual re-training for updates.
- Less adaptable to new query types.

3. Prompt Engineering

Strategy: Template-based clarification prompt.

Example prompt:

”I see you’re having trouble with fractions. Can you tell me:

- Are you working with addition, subtraction, multiplication, or division?
- Is this related to a specific example from your textbook?”

4. Hallucinations and Safety

Definition: Hallucination refers to confident but factually incorrect outputs by LLMs.

Example: Student asks: “What caused the U.S. Civil War?” LLM responds: “It started in 1800 due to the Boston Tea Party.” (Incorrect)

Techniques:

- **Model-level:** Use Retrieval-Augmented Generation (RAG) to ground answers.
- **Post-generation:** Use a verifier model or human-in-the-loop review for factual consistency.

5. Language Model Selection

Trade-offs between 7B and 13B models:

Pros of 13B:

- Greater knowledge capacity.
- Better reasoning and contextual understanding.

Cons of 13B:

- Higher compute and storage cost.
- Increased inference latency.

Conclusion: For K–12 and strict budget/latency requirements, a fine-tuned 7B model may offer the best balance.